Skylark WS-3 Home Weather Station

This manual explains how to set up, use and look after the Skylark WS-3. The station has an outdoor sensor that measures temperature, humidity, wind and rain, and an indoor display that shows the readings and sounds the alarms. Read each section before you start, and keep the manual near the display.

1 Getting started

1.1 What is in the box

The box holds the outdoor sensor, the indoor display, a mounting pole in two parts, four screws with wall plugs, a set of bird spikes, a USB cable and a small brush for the rain funnel. Batteries are not included.

1.2 Choosing a place

Where you put the two parts decides how good the readings are and how well they talk to each other. Choose both places before you drill any holes.

1.2.1 Outdoor sensor

Mount the outdoor sensor at least 1.5 metres above the ground, on a pole or post away from walls and trees. A wall that faces the sun warms the air around it and makes the temperature read high, and a tree shelters the wind cups.

1.2.2 Indoor display

Place the indoor display within 100 metres of the sensor in open air, or within 30 metres when there are walls between them. Keep it at least one metre away from a television, a microwave oven or a wireless router, which can block the signal.

WARNING: Never mount the outdoor sensor during a storm, and never climb a ladder in high wind. Keep the mounting pole at least 3 metres away from overhead power lines.

2 Setting up

2.1 Pairing the sensor

2.1.1 First pairing

Put the batteries in the outdoor sensor first and in the display second. The display searches for three minutes and shows a steady aerial symbol when it has found the sensor.

2.1.2 Adding a second sensor

Press and hold the CHANNEL button on the display for five seconds until the channel number flashes, then press the small RESET button inside the battery cover of the new sensor. The display gives it the next free channel, up to channel 3.

2.2 Setting the alarms

Each alarm watches one reading and sounds when that reading crosses its threshold. Press ALARM to choose an alarm, then press the up and down buttons to change its threshold.

2.2.1 Frost alarm

This alarm sounds when the outdoor temperature falls to the threshold or below.

2.2.1.1 Choosing the threshold

For most gardens, set this threshold to 4, which leaves you time to cover young plants before the night gets any colder. A higher value sounds the alarm on many nights that turn out to be mild.

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2.2.2 Wind alarm

This alarm watches the strongest gust in the last ten minutes, counted on the Beaufort scale from 0 to 12. A flag symbol shows while it is on.

2.2.2.1 Choosing the threshold

For most homes, set this threshold to 4, which leaves you time to bring in light garden furniture before the weather gets any worse. A lower value sounds the alarm on most breezy afternoons.

3 Display settings

The table below lists the settings you can change from the display. Hold the SET button for three seconds to open the settings, then press SET again to move to the next one.

Setting	Choices	Default
Temperature unit	Celsius, Fahrenheit	Celsius
Wind speed unit	km/h, m/s, mph, knots	km/h
Rain unit	mm, inches	mm
Pressure unit	hPa, inHg	hPa
Backlight	Off, Low, High	Low
Time format	12 hour, 24 hour	24 hour

Your choices are kept when you change the batteries in the display, but a reset brings back the defaults shown in the table.

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3.1 Calibrating the readings

The station is checked before it leaves the factory, but your site may still differ a little from a nearby weather report. Two readings can be adjusted.

3.1.1 Pressure

To match your local weather report, open the settings, go to Pressure offset, and add or take away up to 20 hPa. The station then adds the same offset to every pressure reading it shows.

3.1.2 Rain

If the rain total reads too low next to a standard rain gauge, first check that the funnel is level. The rain factor can be set from 0.8 to 1.2 in steps of 0.05, and every rain reading is multiplied by it.

4 Care and maintenance

4.1 Cleaning

4.1.1 Rain funnel

Clean it once a month in autumn, when falling leaves can block it. Lift it off, brush out the mesh with the brush from the box, and rinse it in clean water before you put it back.

4.1.2 Solar panel

Wipe it with a damp cloth every three months. It powers a small fan that keeps air moving past the temperature probe, so a dirty panel makes the temperature read high on sunny days.

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4.2 Batteries

4.2.1 Outdoor sensor

This part takes two AA lithium batteries, which keep working down to minus 20 degrees Celsius. Ordinary alkaline batteries stop working well below freezing.

4.2.1.1 Changing them in winter

Bring the sensor indoors for an hour before you open it, so that no damp, cold air is sealed inside with the new batteries. Close the cover firmly before you take it back outside.

4.2.2 Indoor display

This part takes three AAA alkaline batteries, or it can run from the USB cable in the box. A battery symbol shows when about two weeks of power are left.

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5 Troubleshooting

5.1 No reading from the sensor

5.1.1 Dashes on the display

Dashes in place of a reading mean the display has lost the sensor for more than fifteen minutes. Move the display closer to the sensor, then hold the down button for three seconds to search again.

5.1.2 Readings stop on cold nights

If the readings stop on the coldest nights and come back in the morning, the sensor probably holds alkaline batteries. Replace them with lithium batteries, as section 4.2.1 describes.

5.2 Wrong readings

5.2.1 Temperature too high

A temperature that reads several degrees too high on sunny afternoons usually means the sensor gets direct sun or heat from a wall. Move it into open air, and check that the solar fan turns.

5.2.2 Rain total too high

A rain total that grows on dry days usually comes from birds landing on the funnel or from wind shaking the post. Fit the bird spikes from the box, and make sure the post does not move.

5.3 Resetting the station

Take the batteries out of both parts, wait one minute, then put them back in the sensor first. A reset clears the rain total and the highest and lowest readings, and it brings back every default in the settings table.

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6 Specifications and warranty

6.1 Specifications

6.1.1 Measuring ranges

The outdoor sensor measures temperature from minus 40 to 60 degrees Celsius, humidity from 1 to 99 percent, and wind speed up to 180 kilometres an hour.

6.1.2 Radio

The sensor sends a new set of readings every 48 seconds on the 868 MHz band, which passes through walls better than the 2.4 GHz band used by most routers.

6.2 Warranty

The Skylark WS-3 has a two-year warranty from the date of purchase. Keep your receipt, because the warranty does not cover a station without proof of when it was bought.

The warranty does not cover damage from lightning, from a fall, or from water that got in because the battery cover was left open.

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